



September 2016 Examination

EC400

Probability and Statistical Inference

Instructions to candidates

Time allowed: 2 hours

This paper contains **SEVEN** questions. Answer **ALL** questions.

You are not allowed to use an electronic calculator

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1. **(8 points - 10 minutes)** Let us take a fair coin and flip it 3 times. What is the probability of getting a tail on the third flip given that

- (a) There is a tail on the first flip?
- (b) There is a tail on the the first two flips?
- (c) There is a total of one tail?
- (d) There is a total of two tails?

2. **(8 points - 10 minutes)** Let X and Y be two random variables and $Y = -\sqrt{X+1}$. The probability density function of X is:

$$f_X(x) = \frac{1}{9}x^2, 0 \leq x \leq 3 \\ = 0 \text{ elsewhere.}$$

What is the probability density function of Y ?

3. **(12 points - 15 minutes)** Let (X, Y) have a joint density function

$$f_{X,Y}(x, y) = \begin{cases} 2 & ; \quad 0 < x < y < 1 \\ 0 & ; \quad \text{otherwise} \end{cases}.$$

Derive the unconditional and conditional expectation of Y ($E(Y)$ and $E(Y|X = x)$) and discuss the relation between them.

4. **(12 points - 15 minutes)** Let X and Y be two random variables. Show:
 (X, Y) independent $\Rightarrow Y$ mean independent of $X \Rightarrow X, Y$ uncorrelated.

5. **(18 points - 20 minutes)** Let the random variable X be normally distributed with mean μ and variance 1. You are given a random sample of 100 observations.

- (a) Construct a 95% confidence interval for μ .
- (b) Show that the rejection of the null $H_0 : \mu = 0$ against $H_1 : \mu \neq 0$ with a 5% level of significance, corresponds to the rejection of H_0 when zero does not lie in the 95% confidence interval for μ given in part a).

6. **(25 points - 30 minutes)** Consider a normal distribution of the form $n(\theta, 4)$. You want to test the hypothesis that $H_0 : \theta = 5$ against the alternative $H_1 : \theta > 5$. A random sample $X_1, \dots, X_n$ was obtained.

- (a) Discuss how you would conduct the above test against the one-sided alternative using a 5% level of significance.
- (b) Provide the power of this test at θ_1 where $\theta_1 > 5$. Discuss how the power is affected by the distance of θ_1 from 5 and the sample size. Interpret these findings.
- (c) Specify how you would obtain the power at θ_1 if you had used a two-sided alternative instead with the same level of significance. In general, how does this compare with the power obtained in (b)? Discuss.
- (d) For a given sample of 100 observations, you obtain a point estimate $\bar{x} = 5.2$. Obtain the p-value associated with the one-sided test from (a). Clearly indicate how we should interpret the p-value.

7. **(18 points - 20 minutes)** Let X is a random variable from the Pareto distribution, which has density

$$f(x) = \frac{\theta}{x^{\theta+1}}, \quad x > 1, \quad 1 < \theta < \infty$$

- (a) Obtain the expected value of X .
- (b) A sample of n independent realizations of X ($\{x_1, \dots, x_n\}$) was collected
 - i. Find the maximum likelihood estimator for θ ($\hat{\theta}_{MLE}$).
 - ii. Find the method of moments estimator for θ ($\hat{\theta}_{MME}$).